Task 1:

A researcher wants to predict the life expectancy of countries based on socio-economic and health factors. You are tasked with building a Linear Regression model that can predict life expectancy using historical data.

Dataset: Use the "Life Expectancy (WHO)" dataset from Kaggle (or a similar dataset with numeric features for prediction).

- **Kaggle link:** Life Expectancy (WHO) Dataset
- **Dataset features:**
    - Adult Mortality → mortality rate of adults (per 1000 population)
    - Alcohol → alcohol consumption per capita (litres)
    - GDP → Gross Domestic Product per capita
    - Schooling → average years of schooling
    - HIV/AIDS → prevalence rate (%)
    - Life expectancy → target variable

Linear Regression Task

1. **Load Data:** Import CSV using pandas.
2. **Explore:** Check missing values, basic stats (df.describe()), scatter plots.
3. **Prepare:** Select features (Adult Mortality, Alcohol, GDP, Schooling, HIV/AIDS) and target (Life expectancy), split train/test (80/20).
4. **Build Model:** Use LinearRegression(), fit on training data.
5. **Evaluate:** Predict on test set, calculate **Mean Squared Error (MSE)** and **R^2 score**.
6. **Interpret:** Check model coefficients to see which factors influence life expectancy most, plot predicted vs actual values.

Task 2:

Data set :

https://www.kaggle.com/datasets/prateekmaj21/boston-housing-dataset

A real estate analyst wants to predict the **median value of houses** in different neighborhoods based on various property and neighborhood features. You are tasked with building a **Multiple Linear Regression model** that can predict housing prices using numerical predictors.

- **Dataset features (example):**
  - RM → average number of rooms per dwelling
  - LSTAT → % lower status of the population
  - PTRATIO → pupil-teacher ratio by town
  - INDUS → proportion of non-retail business acres per town
  - NOX → nitric oxides concentration
  - AGE → proportion of owner-occupied units built before 1940
  - MEDV → median value of owner-occupied homes (target variable)

Multiple Linear Regression Task

1. **Load Data:** Import CSV using pandas.
2. **Explore:** Check missing values, stats (df.describe()), scatter plots or correlation heatmap.
3. **Prepare:** Select features (RM, LSTAT, PTRATIO, INDUS, NOX, AGE) and target (MEDV), split train/test (80/20).
4. **Build Model:** Use LinearRegression(), fit on training data.
5. **Evaluate:** Predict, calculate **MSE and R^2**.
6. **Interpret:** Check coefficients to see which features influence house prices most, plot predicted vs actual prices.

Task 3:

A car manufacturer wants to predict the **fuel efficiency (miles per gallon, MPG)** of its vehicles based on **engine size, weight, and horsepower**. You are tasked with building a **Polynomial Regression model** to capture non-linear relationships between features and MPG.

Dataset: Use the “Auto MPG” dataset from Kaggle (or a similar numeric dataset).

- **Kaggle link:** Auto MPG Dataset - Explore Car Performance: Fuel Efficiency Data
- **Dataset features:**
    - Cylinders → number of cylinders in the engine
    - Displacement → engine displacement
    - Horsepower → engine power
    - Weight → vehicle weight
    - Acceleration → time to accelerate from 0–60 mph
    - Model Year → year of manufacture
    - MPG → fuel efficiency (target variable)

Polynomial Regression Task

1. **Load Data:** Import CSV using pandas.
2. **Explore:** Check missing values, descriptive statistics (df.describe()), and scatter plots to observe non-linear relationships.
3. **Prepare:** Select features (Displacement, Horsepower, Weight, Acceleration) and target (MPG), split train/test (80/20).
4. **Transform Features:** Use PolynomialFeatures(degree=2) from sklearn.preprocessing to capture non-linear patterns.
5. **Build Model:** Use LinearRegression() on transformed polynomial features and fit on training data.
6. **Evaluate:** Predict MPG on test set, calculate **Mean Squared Error (MSE)** and **R^2 score**.
7. **Interpret:** Analyze the impact of features, and plot predicted vs actual MPG values to visualize model performance.